

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Performed routine and structured actuarial analyses using **SQL**, **Python**, and **MS Excel** formulas, charts, and graphs to validate healthcare claims experience data, cost assumptions, and expected claims patterns across **PostgreSQL**, **MySQL**, and **Oracle** databases, delivering **95%+ data accuracy** through automated quality checks and reconciliations.
- Developed and built datasets for actuarial modeling teams by designing **ETL pipelines with Python** (pandas, NumPy) and **Power BI** data models, conducting data validation, quality assurance, and reconciliation workflows to ensure compliance with regulatory standards, supporting **8+ concurrent actuarial models** across enterprise systems.
- Supported the development, testing, and maintenance of actuarial models, assumptions, and reports using **R/RStudio** for statistical analysis, regression, and hypothesis testing; applied **machine learning** (supervised/unsupervised algorithms, evaluation metrics) and **predictive modeling** techniques to forecast claims costs and mortality assumptions with **±3% variance tolerance**.
- Collaborated with senior actuaries, analytics teams, and business stakeholders on cross-functional actuarial projects, following defined processes and guidance; created **Power BI dashboards** with **DAX measures** and **Power Query** transformations to communicate actuarial findings, diagnosis codes, procedure codes, and pharmacy terminology insights to non-technical decision-makers.
- Managed healthcare claims data pipelines and reconciliations across interconnected enterprise systems using **REST APIs**, **SQL** stored procedures, and **Python automation scripts**; implemented change-control processes and UAT coordination for model updates and assumption changes, reducing model validation time by **40%**.
- Progressed toward actuarial qualification through structured technical development, mastering **AI and machine learning** in model development, **computer software tools** (Excel, Power BI, R, Python, SAS fundamentals), and healthcare domain knowledge; demonstrated flexibility in learning new tools and handling changing priorities across concurrent actuarial projects and regulatory deadlines.

SKILLS

Actuarial & Analytics Tools – MS Excel (formulas, charts, graphs), Power BI (dashboards, DAX, Power Query), Tableau, R/RStudio, Python (pandas, NumPy, scikit-learn), SAS, machine learning (supervised/unsupervised, evaluation metrics, predictive modeling).

Data Management & Engineering – SQL (PostgreSQL, MySQL, Oracle), data modeling, data warehousing, ETL pipeline development, data validation, quality checks, reconciliation workflows, Hadoop/Spark concepts.

Healthcare Domain – Claims experience, diagnosis codes, procedure codes, pharmacy terminology, expected claims analysis, cost assumptions, mortality assumptions, healthcare data compliance.

Programming & Development – Python, Java, Spring Boot, REST APIs, Flask, FastAPI, Git, CI/CD (Jenkins), Linux, shell scripting.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, containerization, application monitoring (Splunk, Dynatrace).

Professional & Communication – Jira, Confluence, Agile/Scrum ceremonies, cross-functional collaboration, written and verbal communication, change management, UAT coordination, flexibility with changing priorities.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Healthcare Claims Forecasting & Actuarial Model Suite (HCL Technologies)

- Architected and developed an integrated **actuarial analytics platform** combining **Python** data pipelines, **R statistical models**, and **Power BI actuarial dashboards** to support claims cost forecasting, mortality assumption validation, and diagnosis/procedure code analysis for a major healthcare payer client.
- Built **ETL workflows with Python (pandas, NumPy)** to ingest, transform, and reconcile **healthcare claims data** from Oracle and PostgreSQL, applying **data quality checks and validation rules** to ensure **99.2% accuracy** across **2M+ monthly claim records**, reducing manual reconciliation effort by **60%**.
- Developed **predictive machine learning models** using **scikit-learn** (regression, classification) to forecast expected claims costs and identify high-risk populations; integrated **R-based statistical analyses** (hypothesis testing, time-series forecasting) to validate actuarial assumptions, achieving **±2.5% variance** against actual claims experience.
- Designed **Power BI dashboards** with **DAX measures** and multi-level data models to visualize claims trends, diagnosis code distributions, pharmacy cost drivers, and mortality patterns; delivered **8 interactive reports** to senior actuaries and underwriting teams, enabling **real-time assumption reviews** and **quarterly model recalibration**.
- Implemented **automated data quality and reconciliation workflows** using **Python scripts** and **SQL stored procedures** to validate claims accuracy, flag anomalies, and generate audit trails; reduced actuarial model testing cycle from **5 days to 2 days**, supporting faster iteration and regulatory compliance documentation.

SWIFT Payments Operations Dashboard & Regulatory Compliance Automation (HCL Technologies)

- Engineered production-support monitoring systems for a **SWIFT messaging** and payments-operations dashboard serving a Tier-1 banking client, integrating **Splunk** and **Dynatrace** to track transaction flows, message delivery rates, and regulatory compliance metrics in real-time, achieving **99.7% SLA uptime** across **P1/P2 incident response**.
- Developed **Python automation scripts** for payment reconciliation, **ISO 20022 message validation**, and **SEPA/cross-border payments** processing; built data pipelines to extract and harmonize **MT/MX format** payment messages, enabling **automated compliance audit trails** and reducing manual reconciliation by **75%**.
- Collaborated with banking domain business analysts and regulatory teams to implement **GDPR** and data-protection controls, manage change-control procedures for system updates, and coordinate UAT testing for new payment-flow features; delivered **12+ post-implementation production-support engagements** with **zero regulatory findings**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)